

Project SHIELD — Idea Proposal

Formulary & Supply Intelligence

Repository: <https://github.com/lohvinovpavel/MedStockAI>

Problem Statement

- **What problem did you identify?** Hospital pharmacies must find a safe therapeutic substitute the moment a drug enters shortage — today this is a manual, multi-hour search across a formulary of hundreds of SKUs, with no visibility into the risk until the shelf is already empty.
- **Why is it worth solving?** The FDA shortage list has held 100+ active drugs continuously for close to two decades. Each substitution event costs a pharmacist roughly 2–3 hours of manual equivalence research and supplier calls, and an incorrect substitution (same ingredient, wrong indication or dose) is a patient-safety incident, not just an inconvenience.
- **Who benefits?** Hospital pharmacy department (primary user), physicians (fewer blocked orders), hospital finance (lower-cost equivalents), patients (uninterrupted therapy), and the P&T/compliance committee (an audit-ready record of every substitution).

Solution

Product overview

A service that continuously cross-references a hospital's own formulary and stock against public shortage/recall feeds and connected supplier stock, ranks safe lower-cost alternatives with cited justification, and routes every substitution through a licensed pharmacist for sign-off before it reaches a physician's order screen.

User journey

- Exposure Engine cross-references FDA shortage/recall events against the hospital's formulary and burn rate, flagging at-risk drugs with days-of-supply remaining.
- AI ranks safe alternatives (RxNorm equivalence + indication + price) with cited source sentences.
- Pharmacist reviews the ranked list in a queue and approves, edits, or rejects.
- The approved substitution appears as a prompt to the physician at the point of ordering.
- The Pharmacy Director sees a formulary-wide risk and savings dashboard, exportable for compliance.
- Every recommendation, approval, and override is written to an append-only audit log.

AI capabilities

- Equivalence reasoning over the RxNorm graph conditioned on indication — not a static lookup table.
- Unstructured-text extraction from shortage/label notices, with mandatory exact-sentence citation.
- Supplier Connector Factory: the LLM proposes a declarative mapping spec (not executable code) from a sample distributor payload to a canonical stock schema, once per supplier, confirmed by a human before use.
- Self-healing on schema drift: the LLM proposes a spec diff when a supplier silently changes format; a human approves before it takes effect.
- A ~30-case golden-set evaluation runs in CI to catch AI quality regressions.

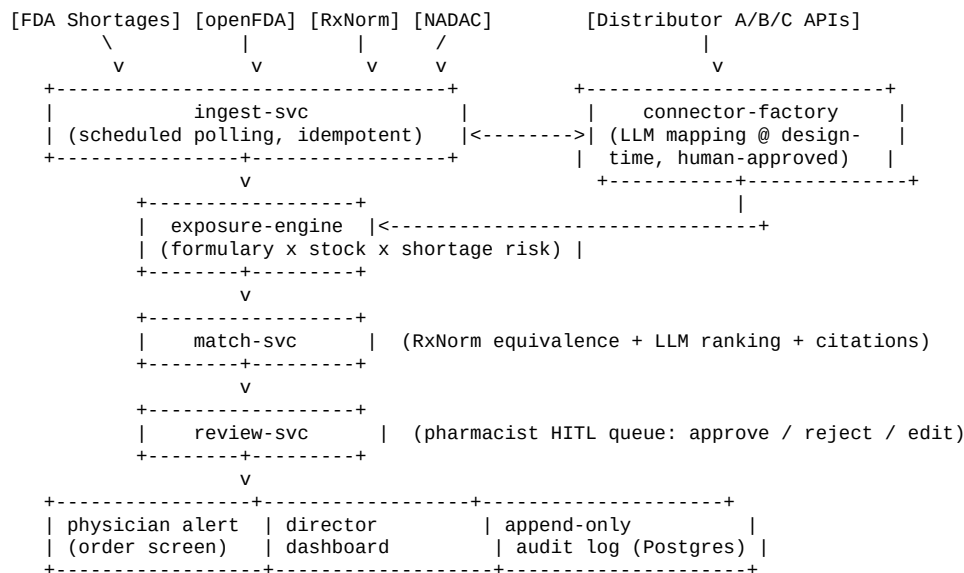
Live integrations

- FDA Drug Shortages Database — public API, no key required
- openFDA — Enforcement Reports / Drug Recalls / Drug Label — public API, no key required
- RxNorm / NLM — drug equivalence graph — public API, no key required
- CMS NADAC — public reference drug pricing — no key required

All four are free and keyless, reducing the single-point-of-failure risk on the assignment's “at least one live integration” requirement.

Architecture

High-Level Architecture Diagram



ER Diagram (entities & relationships)

Entities: Drug(rxcul, ndc, name) · FormularyItem(hospital_id, drug_id, reorder_threshold) · StockSnapshot(formulary_item_id, qty, observed_at, source) · ShortageEvent(drug_id, source, detected_at, raw_text) · SupplierFeed(id, distributor, schema_version) · MappingSpec(supplier_feed_id, version, spec_json, approved_by) · SubstitutionRecommendation(shortage_event_id, candidate_drug_id, rank, citation, cost_delta) · ReviewDecision(recommendation_id, pharmacist_id, decision, timestamp) · AuditLogEntry(actor, action, payload, timestamp — append-only, no update/delete).

Relationships: Hospital 1—* FormularyItem · FormularyItem 1—* StockSnapshot · Drug 1—* ShortageEvent · ShortageEvent 1—* SubstitutionRecommendation · SubstitutionRecommendation 1—1 ReviewDecision · every state-changing action → AuditLogEntry.

Key technology decisions

- FastAPI + PostgreSQL, one Docker service per pipeline stage — single-responsibility, independently deployable and scalable.
- The LLM is used only where reasoning over unstructured or heterogeneous input is required (ranking, extraction, schema mapping). Runtime data transforms run through a deterministic declarative interpreter — never LLM-generated code — removing a non-determinism and injection risk.
- A canonical StockItem model decouples supplier-specific formats from the rest of the system, so adding a new supplier never touches downstream services.

Deployment approach

docker-compose for the MVP/demo. The same service boundaries map directly onto a container orchestrator (Kubernetes/ECS) for production — each service scales independently by its own load profile (ingest is I/O-bound, match-svc is LLM-call-bound).

Security considerations

- Role-based access control per persona (pharmacist / physician / director).
- No patient-identifiable data is required by this scope — only formulary lines and hospital-side stock counts.
- Secrets (API keys, DB credentials) live in environment/secret manager, never in mapping specs or logs.
- Append-only audit log — no update/delete path is exposed to any service.
- Mapping specs are validated against a fixed schema and an allow-listed transform set; the interpreter never executes model-authored code.

Business Impact & Differentiation

Expected business impact

- Days of advance warning instead of an empty-shelf event.
- ~2–3 pharmacist-hours saved per substitution event.
- Measurable savings from cheaper equivalent selection (price delta × substitution volume).
- Ready-made audit trail for Joint Commission / P&T reviews.

Scalability

Per-supplier integration cost drops from a hand-written adapter (days of engineering) to a human confirming a machine-proposed spec (hours) — this is what makes 10-site and 100-site rollouts economically different from a linear cost curve.

Competitive advantage (Moat)

The combination of (a) a hospital-specific risk view no public feed offers alone, and (b) an accumulating library of approved supplier mapping specs, is hard for a generic drug-database lookup tool to replicate — every additional customer or supplier makes onboarding the next one cheaper.

Future roadmap

Recall/label-change monitoring on the same pipeline; expiry-based waste tracking; GPO/340B contract-pricing integration; patient-facing transparency on substitutions.